PassFrame tuition

Measurement: Length, Mass & Capacity

Year 5 Accelerated — Homework ⏱️ 45 min

Worked example — read this through before you start

Worked example: A tank holds 36 litres and is drained at 2.4 litres per minute. How long does it take to empty completely?

Method & answer: 36 ÷ 2.4 = 15 minutes.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Convert 4.25kg to grams.

2. Convert 8,750m to km.

3. Convert 0.045km to metres. A) 4.5m B) 45m C) 450m D) 4,500m

4. Convert 3.6l to ml.

5. Convert 5480g to kg.

6. Select ALL of these that equal 3.25 litres: A) 3,250ml B) 3l + 250ml C) 3,025ml D) 13 × 250ml E) 32.5 × 100ml

7. Add: 2.7m + 3.65m. Give your answer in metres.

8. Subtract: 7.4kg − 2.85kg.

9. Which of these is equal to 0.8km? A) 80m B) 800m C) 8,000m D) 0.08m

10. A tank holds 30 litres. Water is used at a rate of 4 litres per hour. How many litres remain after 5 hours?

11. Order from smallest to largest: 3.2kg, 3,150g, 3.25kg.

12. Select ALL of these masses that are heavier than 4.5kg: A) 4,450g B) 4.55kg C) 4,600g D) 4.4kg E) 4,520g

13. A ribbon 16.8m long is cut into pieces of 1.2m each. How many complete pieces can be cut?

14. A jug holds 2.4l. If 850ml is poured out, how much remains? A) 1.45l B) 1.55l C) 1.65l D) 1.75l

15. A cable is 9.6m long and is cut into pieces of 45cm each. How many complete pieces can be cut, and how much is left over?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A runner travels 5.2km, then a further 3,600m. What is the total distance in km?

2. A recipe needs 0.4kg of butter per batch. How much is needed for 7 batches, in kg?

6. Select ALL of these that could combine to make exactly 4 litres: A) 2l + 2l B) 1,600ml + 2,400ml C) 3l + 900ml D) 8 × 500ml E) 3,500ml + 400ml

7. A gardener has 22m of fencing. He uses 8.4m and 7.6m for two sides. How much fencing does he have left?

11. Select ALL of these weights that round to 5.8kg to 1 decimal place: A) 5.75kg B) 5.84kg C) 5.749kg D) 5.83kg E) 5.799kg

12. A tank contains 72 litres. It is filled at 6 litres per minute. How long does it take to fill, in minutes?

3. A tank holds 54 litres and is drained at 4.5 litres per minute. How long does it take to empty? A) 10 min B) 11 min C) 12 min D) 13 min

4. Three parcels weigh 3.4kg, 2,650g and 1.95kg. What is their total weight in kg?

5. A plank of wood is 5.4m long. Six shelves of 85cm each are cut from it. How much wood is left, in cm?

8. A recipe for 1 loaf needs 320g of flour. A baker has 4kg of flour. How many whole loaves can be made, and how much flour is left over?

9. A car's fuel tank holds 60 litres. It uses fuel at 7.5 litres per 100km. How many litres are used on a 220km journey? A) 15.5l B) 16.0l C) 16.5l D) 17.0l

10. A ribbon 18m long is cut into pieces 1.5m long. How many complete pieces can be cut?

13. A suitcase weighs 17.6kg. Airline limit is 23kg. How much more weight can be added before the limit is reached?

14. A recipe scales up: 1 batch needs 0.35l of milk. How much is needed for 6 batches? A) 1.8l B) 2.0l C) 2.1l D) 2.4l

15. A tank is filled with water at 3 litres per minute for 10 minutes, then drained at 2.5 litres per minute for 6 minutes. How much water is in the tank at the end?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A tank holds 96 litres. It is filled at 8 litres per minute for the first 6 minutes, then at 6 litres per minute after that. How long in total does it take to fill the tank?

2. A metal rod 3.5m long expands by 0.8% when heated. What is its new length, to the nearest mm? A) 3,514mm B) 3,528mm C) 3,540mm D) 3,550mm

3. A recipe for 8 people needs 1.2kg of rice. How much rice is needed for 20 people, keeping the ratio the same?

4. Select ALL of these that are within 30g of 3.5kg: A) 3,480g B) 3,460g C) 3,525g D) 3,540g E) 3,470g

5. A runner covers 4.8km every 16 minutes at a steady pace. How far do they travel in 45 minutes at the same rate?

6. A water tank holds 220 litres. It loses water through a leak at 0.85 litres per hour. How many litres remain after 3 full days (72 hours)?

7. A ribbon of length 8.4m is divided in the ratio 3:4. What is the length of the shorter piece? A) 3.2m B) 3.4m C) 3.6m D) 3.8m

8. A rectangular tank has a capacity of 42,000 litres and is filled at 700 litres per minute. How many hours does it take to fill?

9. Four sacks of flour weigh 3.2kg, 4.75kg, 2.85kg and 1.2kg. If the flour is repackaged equally into 6 new bags, how much does each new bag weigh?

10. A recipe scales in proportion to the number of people fed: 500g feeds 5 people. Select ALL of the statements below that correctly describe scaling this up or down: A) For 8 people, use 800g B) For 3 people, use 300g C) For 12 people, use 1.2kg D) For 2 people, use 150g E) For 15 people, use 1.5kg

11. A tank contains 65 litres of water. 40% is drained out. How many litres remain?

12. A plank of wood 4.8m long is cut so that one piece is three times as long as the other. Find the length of the shorter piece, in cm.

13. A lorry can carry a maximum load of 4.5 tonnes. It is already carrying 3,950kg. How many more 20kg bags can it carry? A) 25 B) 27 C) 28 D) 30

14. A recipe uses flour, sugar and butter in the ratio 7:4:3 by weight. If the total weight is 2.1kg, how much sugar is used, in grams?

15. A water tank is 45% full and contains 108 litres. What is the full capacity of the tank, in litres?